



2025

C23

BOOKLET NO.

174001

Statistics

Paper – I

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.



SECTION – A

Q1. Solve the following.

(5×10=50)

Q1. (a) Three urns of same appearance have the following proportions of balls.

V_1 – 2 white, 1 black ; V_2 – 1 black, 2 white ; V_3 – 2 white, 2 black. One of the urn is selected and one ball is drawn. It turns out to be white. What is the probability of drawing white ball again, when the ball drawn at first draw is not replaced ?

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Q1. (b) Two dice are tossed. Find the probability that

- 1) A : Even number on first dice
- 2) B : Sum of the numbers on two dice is 8
- 3) A or B
- 4) A and B

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Q1. (c) A random sample $(X_1, X_2, X_3, X_4, X_5)$ of size 5 is drawn from a normal

population with unknown mean μ , consider the following estimators to estimate

$$\mu, t_1 = \frac{X_1 + X_2 + X_3 + X_4 + X_5}{5}, t_2 = \frac{X_1 + X_2}{2} + X_3 \text{ and } t_3 = \frac{2X_1 + X_2 + \lambda X_3}{3}, \text{ where}$$

λ is such that, t_3 is an unbiased estimator of μ . Find λ . Are t_1 and t_2 unbiased ?

State giving the reasons, the estimator which is best among t_1 , t_2 and t_3 .

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Q1. (d) Show that $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ in random sampling from $f(x, \theta) = \begin{cases} \frac{1}{\theta} \exp(-x/\theta) & 0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$

where $0 < \theta < \infty$, is an MVB estimator of θ and has variance θ^2/n .

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Q1. (e) In random sampling from normal population $N(\mu, \sigma^2)$, find the maximum likelihood estimator for

- i) μ when σ^2 is known, and
- ii) σ^2 when μ is known

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Q2. (a) 1) For below given bivariate probability distribution of X and Y find

i) $P(X \leq 1, Y = 2)$

ii) $P(X \leq 1)$

iii) $P(Y \leq 3)$

iv) $P(X < 3, Y \leq 4)$

v) $P(X = 2, Y \geq 4)$

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X \ Y	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

Q2. (a) 2) A random variable X has the following probability distribution.

Values of X	0	1	2	3	4	5	6	7
P(X)	0	K	2K	2K	3K	K^2	$2K^2$	$7K^2 + K$

i) Find K

ii) Evaluate $P(X \leq 6)$ and $P(0 < X < 5)$

iii) If $P(X \leq a) > \frac{1}{2}$ find maximum value of 'a'.

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Q2. (b) A random variable X takes the values of 0, 1, 2 with respective probabilities $\frac{\theta}{4N} + \frac{1}{2} \left(1 - \frac{\theta}{N}\right)$, $\frac{\theta}{2N} + \alpha \left(1 - \frac{\theta}{N}\right)$ and $\frac{\theta}{4N} + \left(\frac{1-\alpha}{2}\right) \left(1 - \frac{\theta}{N}\right)$, where N is known

number and α & θ are unknown parameters. If 75 independent observations on X yielded the values of 0, 1, 2 with frequencies 27, 38, 10 respectively, estimate θ and α by method of moments.

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Q2. (c) Examine whether a best critical region exists for testing the null hypothesis $H_0 : \theta = \theta_0$ against alternative hypothesis $H_1 : \theta = \theta_1$ for the parameter θ of

the distribution $f(x, \theta) = \frac{1+\theta}{(x+\theta)^2} \quad 1 \leq x < \infty$.

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- Q3. (a)** The life of gears in weeks of certain machine is continuous random variable with
- p, d, f is given by $f(x) = \begin{cases} 100/x^2 & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$
- What is probability that all of three gears will have to replace during first 150 weeks ?
 - What is the probability that none of the gear will have to replace during first 150 weeks ?
 - What is the probability that gear will last less than zero weeks, if it is known that gear is working after 150 weeks ?
 - What is maximum number of gears that may be inserted into machine so that there is probability of 0.5 that after 150 weeks of service all of them are still working. 20

- Q3. (b)** Following table gives the probabilities and observed frequencies in four classes AB, Ab, aB, ab in genetical experiment. Examine parameter θ by method of maximum likelihood estimation

Class	AB	Ab	aB	ab
Probability	$(2 + \theta)/4$	$(1 - \theta)/4$	$(1 - \theta)/4$	$\theta/4$
Frequency	108	27	30	8

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- Q3. (c)** If $X_1, X_2, \dots, X_n$ is random sample from $N(\mu, \sigma^2)$ population. Find sufficient estimators for μ and σ^2 . 10

- Q4. (a)** Let $X \sim N(\mu, 4)$, μ unknown, to test $H_0 : \mu = -1$ against $H_1 : \mu = 1$ based on sample of size 10 from this population, we use the critical region $x_1 + 2x_2 + \dots + 10x_{10} \geq 0$. What is its size ? What is the power of the test ? 20

- Q4. (b)** Show that for the distribution, $dF(x) = \theta \exp(-x\theta)$ $0 < x < \infty$ critical confidence limits for the large samples with 95% confidence coefficient are given by
- $$\theta = \left(1 \pm \frac{1.96}{\sqrt{n}} \right) / \bar{x}.$$
- 20

- Q4. (c)** If X is the number scored in a throw of a fair dice, show that the Chebychev's inequality gives, $P(|x - \mu| > 2.5) < 0.47$ where μ is the mean of X , while the actual probability is zero. 10



SECTION – B

Q5. Solve the following. (5×10=50)

Q5. (a) If X and Y are random variables and a, b, c and d are any constants such that $a \neq 0$ and $c \neq 0$, then prove that $r(aX + b, cY + d) = \frac{ac}{|ac|} rXY$. 10

Q5. (b) What do you understand by discriminant analysis and principal component analysis. Explain their purpose. 10

Q5. (c) Agronomist wishes to study response of nitrogen and phosphorus on yield of wheat by using three levels of each with four replications. Suggest suitable design give its layout and ANOVA. 10

Q5. (d) Explain the conditions under which cluster sampling and two stage sampling methods are used. Explain with two examples of each. 10

Q5. (e) What are the principles and advantages of stratified random sampling ? 10

Q6. (a) For trivariate population with X_1, X_2 and X_3 following results were obtained

Means : $\bar{X}_1 = 10, \bar{X}_2 = 15, \bar{X}_3 = 20$

Variances : $V(X_1) = 4, V(X_2) = 9, V(X_3) = 16$ and

Correlations : $r_{12} = 0.2, r_{13} = 0.4, r_{23} = 0.6$

i) Fit regression plane $X_1 = a + b_1X_2 + b_2X_3$ and

ii) Estimate X_1 when $X_2 = 10$ and $X_3 = 25$ 20

Q6. (b) In an experiment with three fertilizers N, P and K each at two levels 0 and 1 each the yield data (kg/plot) due to eight treatment combinations of RBD with three replicates is given below. Calculate main effects and interaction sum of squares by Yule's method. 20

Treat	R – I	R – II	R – III
(1)	40	53	38
n	35	59	18
p	40	69	43
np	64	78	52
k	71	72	58
nk	26	31	19
pk	43	45	27
npk	51	62	40



- Q6.** (c) For studying milk yield, feeding and management practices followed by the farmers of milch animals in year 1977 – 78, the state of Haryana was divided in to 4 zones according to agro-climatic conditions. The total number of milch animals in 17 randomly selected villages (in 1977 – 78) of zone A (Y) along with their livestock census data in 1976 (X) are shown below.

S. N.	Y	X
1	1129	1141
2	1144	1144
3	1125	1127
4	1138	1153
5	1137	1117
6	1127	1140
7	1163	1153
8	1153	1146
9	1164	1189
10	1130	1137
11	1153	1170
12	1125	1115
13	1116	1130
14	1115	1118
15	1112	1122
16	1112	1113
17	1123	1166

X : Number of milch animals in census

Y : Number of milch animals in survey

Estimate the total number of milch animals in 11Y villages of zone A by ratio method of estimation, when the total number of milch animals in the census = 143968.



- Q7. (a)** How will you fit regression line $y = a + bx$ to bivariate data ? Extend the procedure to fit exponential curve $y = ab^x$. **20**
- Q7. (b)** The number of pepper standards for selected villages in each of the three strata of Trivandrum zone is as follows.

Stratum No.	Total number of villages in the stratum	Number of villages selected from the stratum	Number of pepper standards in each of selected villages
1	441	11	41, 116, 19, 15, 144, 159, 212, 57, 28, 119, 76
2	405	12	39, 70, 38, 37, 161, 38, 27, 119, 36, 128, 30, 208
3	103	7	252, 385, 192, 296, 115, 159, 120

Estimate the total of pepper standards along with its standard error. **20**

- Q7. (c)** Research and development manager of an automobile company wishes to study the effect of the tyre brand on the tread loss (in mm) of tyre.
- Four different tyres of four different brands (A, B, C and D) are fitted at four different positions of the four cars because there might be variability from one car to another car as well as from one position to another position. The data as per design are as given below.

Positions ↓	Cars			
	C ₁	C ₂	C ₃	C ₄
P ₁	A	B	C	D
	8	7	1	7
P ₂	B	C	D	A
	11	2	7	8
P ₃	C	D	A	B
	6	8	6	6
P ₄	D	A	B	C
	4	2	3	8

Analyse the data and give your comments.

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- Q8.** (a) Explain the concept of T^2 -statistic and prove that likelihood ratio test of the hypothesis $H_0 : \mu = \mu_0$ is $T^2 \geq T_0^2$

where $T_0^2 = (N-1) \left(\lambda_0^{-2/N} - 1 \right)$ and

$$\lambda_0 = \frac{|A|}{1 + \left[\sqrt{N} (\bar{X} - \mu_0) \right] \left[\sqrt{N} (\bar{X} - \mu_0) \right]^T}$$

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- Q8.** (b) What do you understand by confounding in 2^3 factorial experiments. Give plan and ANOVA for three replicates of complete confounded design in which highest order interaction is confounded.

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- Q8.** (c) Explain the principles and advantages of sampling over census.

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